WASP-43b

R-0477

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-43_231212` · created 2026-07-20T05:01:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460290.9492 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.009 +/- 0.054
Transit depth (Rp/Rs)^2	0.0078 +/- 0.096 [%]
Orbital Inclination (inc)	77.16 +/- 2.92
Airmass coefficient 1 (a1)	0.82 +/- 0.11
Airmass coefficient 2 (a2)	0.102 +/- 0.08
Scatter in the residuals of the lightcurve fit is	12.95 %
Best Comparison Star	#3 - [262, 412]
Optimal Aperture	1.77
Optimal Annulus	5.8
Transit Duration (day)	0.012 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.009 ± 0.054 vs 0.1594 (deviation 2.79σ)
Duration fit vs published	0.012 d vs 0.0512 d (deviation 2.45σ)
Mid-time O–C	4.8 min
Depth SNR	0.2
Residual scatter	12.95 %

Light curve

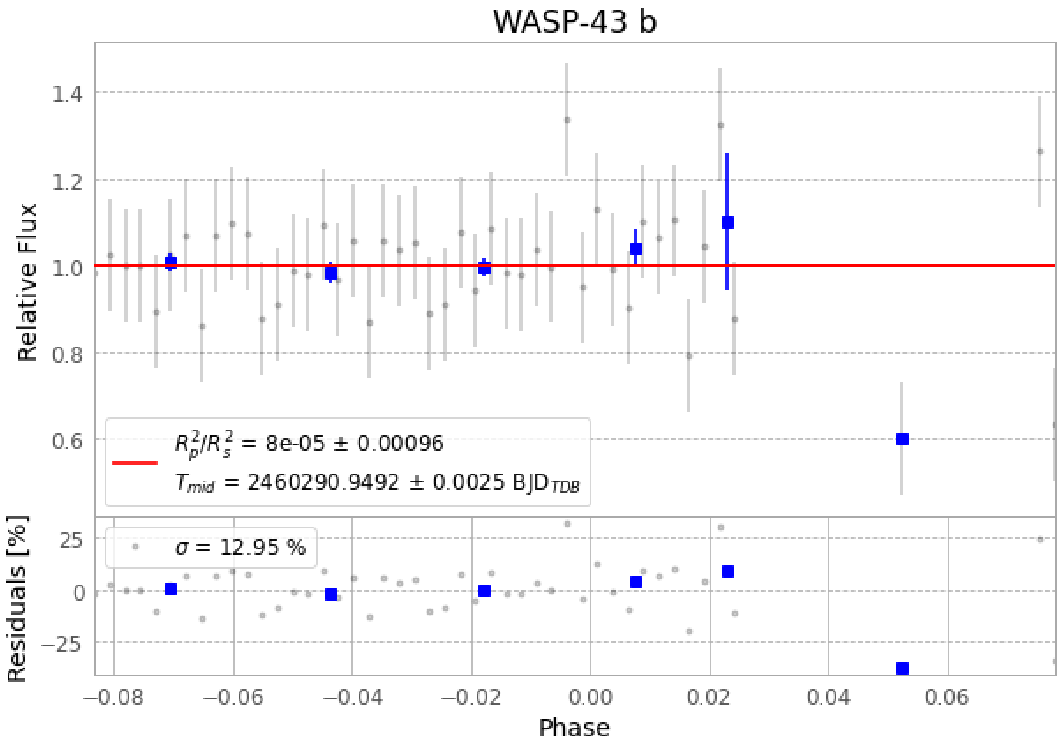

Figure 1 — FinalLightCurve_WASP-43 b_2023-12-12.png (SHA-256 ee2b4fdee7fd8002…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [426, 233], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 2ef54ab43fe45dab…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80`

Data provenance

64 FITS frames (42 MB), `WASP-43231212090617.FITS` → `WASP-43231212121515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-231212.log (6ddaa06b4a62…), FinalLightCurve_WASP-43 b_2023-12-12.png (ee2b4fdee7fd…), FinalLightCurve_WASP-43 b_2023-12-12.csv (146c6e3ee3a0…)

Compute resources

Wall time 320.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-20 05:01Z.